

JIAYI (ERIS) ZHANG

<https://eriszhang.github.io> ♦ eriszhan@stanford.edu ♦ +1 650 683 2633

Computing and Data Science (CoDa), Room E308 - 389 Jane Stanford Way, Stanford, CA 94305

Drawing on my expertise in physics simulation, geometry processing, numerical optimization and spatial intelligence, I develop algorithms, models, and tools that enhance human productivity and creativity in science, engineering, and creative authoring workflows. Focusing on level-of-detail simulation, I create multiscale methods that deliver fast, predictive previews with high-fidelity physics, enabling simulators that balance speed, accuracy, and control for scalable real-world use.

EDUCATION

Stanford University

Doctor of Philosophy Student in Computer Science

Sept. 2021 - Ongoing

Stanford, CA

- NVIDIA Graduate Fellowship (10 worldwide selected from 600+ applicants)
- Roblox Graduate Fellowship (2 worldwide)
- Stanford Graduate Fellowship - Reed-Hodgson Fellow
(Selected 7 from 120+ Stanford CS Ph.D. admits; the only 1 in Computer Graphics)
- MIT EECS Rising Star & WiGRAPH Rising Star
- Heidelberg Laureate Forum Young Researcher

Advisor: Prof. Doug L. James

University of Toronto

Honors BSc in Computer Science and Mathematics

Sept. 2016 - June 2021

Toronto, ON

Advisor: Prof. Alec Jacobson *Overall GPA: 3.93/4.0*

PROFESSIONAL EXPERIENCE AND INTERNSHIPS

NVIDIA

Research Intern, High-Fidelity Physics Group with Dr. Ken Museth

June 2025 - Ongoing

Santa Clara, CA

- Topic: GPU-Based MPM Simulation

Adobe Research

Research Intern, MOVIN Lab with Dr. Danny Kaufman

June 2024 - Sept. 2024

Seattle, WA

- Topic: Progressive Volumetric Elastodynamics

Adobe Research

Research Intern, BIG Lab with Dr. Danny Kaufman

June 2023 - Sept. 2023

Seattle, WA

- Topic: Progressive Dynamics for Animation Authoring

Adobe Research

Research Intern, Creative Intelligence Lab with Dr. Danny Kaufman

June 2022 - Sept. 2022

Seattle, WA (Remote)

- Topic: Progressive Shell Quasistatics

Adobe Research

Research Intern, Creative Intelligence Lab with Dr. Danny Kaufman

June 2021 - Sept. 2021

Seattle, WA (Remote)

- Topic: Progressive Cloth Quasistatics

Adobe Research

Research Intern, Emerging Graphics Group with Dr. Qi Sun

June 2020 - Sept. 2020

San Jose, CA (Remote)

- Topic: Human Skin Modeling for Efficient Deformation

DGP Lab, University of Toronto

Research Assistant with Profs. David I.W. Levin and Alec Jacobson

Sept. 2019 - May 2020

Toronto, ON

- Topic: Secondary Physical Motion for Rig-Based Animation

DGP Lab, University of Toronto

Research Assistant with Profs. Marc Alexa and Alec Jacobson

Mar. 2019 - August 2019

Toronto, ON

- Topic: Efficient Computation for the Orthogonal Procrustes Problem

DGP Lab, University of Toronto

Research Assistant with Prof. Fanny Chevalier

Apr. 2018 - Sept. 2019

Toronto, ON

- Topic: Image-Editing-Driven Design Tool for Pictorial Visualizations

Numerical Analysis Group, University of Toronto

Research Assistant with Prof. Kenneth R. Jackson

May 2018 - Sept. 2019

Toronto, ON

- Topic: Two-Level Importance Sampling Method for Financial Portfolio Credit Risk Simulation

PUBLICATIONS

1. **Progressing Level-of-Detail Animation of Volumetric Elastodynamics**
Jiayi Eris Zhang, Doug L. James, Danny M. Kaufman
 Preprint, 2025, <https://arxiv.org/abs/2509.14177>
2. **Progressive Dynamics++: A Framework for Stable, Continuous, and Consistent Animation Across Resolution and Time**
Jiayi Eris Zhang, Doug L. James, Danny M. Kaufman
 ACM Transactions on Graphics (SIGGRAPH 2025)
3. **Level-of-Detail for Geometry Processing and Simulation**
Jiayi Eris Zhang, Hsueh-Ti Derek Liu
 ACM SIGGRAPH Course 2025
4. **Progressive Dynamics for Cloth and Shell Animation**
Jiayi Eris Zhang, Doug L. James, Danny M. Kaufman
 ACM Transactions on Graphics (SIGGRAPH 2024)
Featured in the technical papers trailer
Selected for the front cover of the TOG journal proceedings
5. **Progressive Shell Quasistatics for Unstructured Meshes**
Jiayi Eris Zhang, Jérémie Dumas, Raymond Yun Fei, Alec Jacobson, Doug L. James, Danny M. Kaufman
 ACM Transactions on Graphics (SIGGRAPH Asia 2023)
6. **Fast Complementary Dynamics via Skinning Eigenmodes**
 Otman Benckekroun, **Jiayi Eris Zhang**, Siddhartha Chaudhuri, Eitan Grinspun, Yi Zhou, Alec

Jacobson
ACM Transactions on Graphics (SIGGRAPH 2023)

7. **Progressive Simulation for Cloth Quasistatics**

Jiayi Eris Zhang, Jérémie Dumas, Raymond Yun Fei, Alec Jacobson, Doug L. James, Danny M. Kaufman

ACM Transactions on Graphics (SIGGRAPH Asia 2022)

Selected for the back cover of the TOG journal proceedings

8. **Surface Multigrid via Intrinsic Prolongation**

Hsueh-Ti Derek Liu, **Jiayi Eris Zhang**, Mirela Ben-Chen, Alec Jacobson

ACM Transactions on Graphics (SIGGRAPH 2021)

9. **Fast Updates for Least-Squares Rotational Alignment**

Jiayi Eris Zhang, Alec Jacobson, Marc Alexa

Computer Graphics Forum (Eurographics 2021)

10. **Complementary Dynamics**

Jiayi Eris Zhang, Seungbae Bang, David I.W. Levin, Alec Jacobson

ACM Transactions on Graphics (SIGGRAPH Asia 2020)

Featured in the technical papers trailer and Two Minute Papers

11. **DataQuilt: Extracting Visual Elements from Images to Craft Pictorial Visualizations**

Jiayi Eris Zhang, Nicole Sultanum, Anastasia Bezerianos, Fanny Chevalier

ACM CHI Conference on Human Factors in Computing Systems (CHI 2020)

HONORS AND AWARDS

NVIDIA Graduate Fellowship	2025
-----------------------------------	------

Awarded to 10 outstanding graduate students (out of 600+ applicants) worldwide in AI.

WiGRAPH Rising Star	2025
----------------------------	------

Jane Street Graduate Fellowship Finalist	2025
---	------

Declined to proceed upon receiving another industry fellowship.

Roblox Graduate Fellowship	2024
-----------------------------------	------

Awarded to 2 outstanding graduate students worldwide in scalable computing and related fields.

EECS Rising Star by EECS at MIT	2024
--	------

Young Researcher for the 11th Heidelberg Laureate Forum	2024
--	------

Snap Research PhD Fellowship Finalist	2022
--	------

Stanford Graduate Fellowship - Reed-Hodgson Fellow	2021
---	------

Awarded to fully fund outstanding doctoral students for 12 quarters.

Adobe Research Women-in-Technology Scholarship	2020
---	------

Awarded to outstanding female undergraduate/master computer science students worldwide.

CRA Outstanding Undergraduate Researchers Award Finalist	2020
---	------

Awarded to top undergraduate computer science researchers in North America.

University of Toronto Excellence Award UTEA	2019
--	------

Dean's Honor List	2017 - 2021
--------------------------	-------------

George Luste Prize in 1st Year Physics	2018
---	------

George Gray Falle Scholarship	2017
University of Toronto Scholar	2017
University of Toronto Admission Scholarship	2016

PATENT

Improved Modeling Interfaces with Progressive Cloth Simulation

Daniel Kaufman, Yun Fei, Jeremie Dumas, Alec Jacobson, **Jiayi Zhang**

· US Patent No.: 058083-1330670

INVITED TALKS

Progressive Dynamics++: A Framework for Stable, Continuous, and Consistent Animation Across Resolution and Time

Technical paper presentation at SIGGRAPH 2025

August 2025
Vancouver, Canada

Progressive Simulation for Scalable Experiences

Invited lightning talk speaker at SIGGRAPH ROM workshop

August 2025
Vancouver, Canada

Progressive Simulation for Scalable Experiences

Research talk at Stanford's Graphics Café Talk Series

May 2025
Stanford, USA

Progressive Dynamics for Cloth and Shell Animation

Research talk at Roblox's research seminar hosted by Dr. Victor Zordan

March 2025
Virtual

Progressive Dynamics for Cloth and Shell Animation

Research talk at the VCG Summer School at Peking University hosted by Prof. Mengyu Chu

August 2024

Virtual

Progressive Dynamics for Cloth and Shell Animation

Research talk at NVIDIA's high-fidelity physics group hosted by Prof. David I.W. Levin

August 2024
Virtual

Progressive Dynamics for Cloth and Shell Animation

Technical paper presentation at SIGGRAPH 2024

August 2024
Denver, USA

Progressive Shell Quasistatics for Unstructured Meshes

Technical paper presentation at SIGGRAPH ASIA 2023

December 2023
Sydney, Australia

Progressive Simulation for Cloth Quasistatics

Research talk at Peking University hosted by Dr. Bin Wang

March 2023
Beijing, China

Progressive Simulation for Cloth Quasistatics

Research talk at the Graphics And Mixed Environment Seminar organized by MSRA

March 2023
Virtual

Progressive Simulation for Cloth and Shells

Research talk at the University of Toronto hosted by Prof. Alec Jacobson

March 2023
Toronto, Canada

Progressive Simulation for Cloth Quasistatics

Technical paper presentation at SIGGRAPH ASIA 2022

December 2022
Daegu, South Korea

Complementary Dynamics

Technical paper presentation at SIGGRAPH ASIA 2020

December 2020
Daegu, South Korea (Virtual)

Complementary Dynamics

Research talk at the Pixar Animation Studios hosted by Dr. Fernando de Goes

November 2020
Virtual

Complementary Dynamics	November 2020
Research talk at the Massachusetts Institute of Technology hosted by Prof. Justin Solomon	Virtual
Complementary Dynamics	November 2020
Research talk at the Epic Games hosted by Dr. Ryan Schmidt	Virtual
Complementary Dynamics	November 2020
Research talk at the Graphics And Mixed Environment Seminar organized by MSRA	Virtual
Complementary Dynamics	October 2020
Opener talk for Dr. Danny Kaufman at the Toronto Geometry Colloquium	Virtual
DataQuilt: Extracting Visual Elements to Craft Pictorial Visualizations	May 2020
Technical paper presentation at CHI 2020	Honolulu, USA (Virtual)
Exterior Rig Space	December 2019
Research talk at the Montreal-Toronto pre-SIGGRAPH workshop	Waterloo, Canada
Expressive Design for Infographics Authoring	September 2018
Undergraduate Research in Computer Science Conference	Toronto, Canada

TEACHING EXPERIENCE

Level-of-Detail for Geometry Processing and Simulation	July 2024
Lecturer for a SIGGRAPH course co-taught with Dr. Hsueh-Ti Derek Liu	
Introduction to Elastodynamic Simulation	August 2025
Invited lecturer at the Graduate School of Symposium of Computer Animation (SCA) 2025	
CS 248B Fundamentals of Computer Graphics: Animation and Simulation	Fall 2024
Teaching Assistant with Profs. C. Karen Liu and Doug James	
Introduction to Elastic Simulation for Solids, Cloth and Rods	August 2024
Invited lecturer at the Graduate School of Symposium of Computer Animation (SCA) 2024	
Level of Detail (LOD) for Geometry Processing and Simulation	July 2024
Tutorial lecturer at the Summer Geometry Initiative (SGI) 2024 hosted by Prof. Justin Solomon	
Introduction to Optimization Time Integration for Solids and Fluids	June 2024
Invited lecturer at the Graduate School of Symposium of Geometry Processing (SGP) 2024	
CS 348C Computer Graphics: Animation and Simulation	March 2023, 2024, 2025
Guest lecturer on topics of contact and progressive simulation hosted by Prof. Doug James	
Skinning: Real-time Shape Deformation	July 2022, 2023
Tutorial lecturer at the Summer Geometry Initiative (SGI) hosted by Prof. Justin Solomon	
Physically Based Shape Deformation	July 2021
Tutorial lecturer at the Summer Geometry Initiative (SGI) hosted by Prof. Justin Solomon	
CSC419/2520 Geometry Processing	Fall 2020
Teaching Assistant with Prof. Alec Jacobson	

MENTORSHIP EXPERIENCE

Yuchen Su

June 2025 - Ongoing

Undergraduate Student (UGVR), Tsinghua University

Topic: Progressive Sound Synthesis

Qihong (Jimmy) Chen

October 2024 - Ongoing

Master's Student, Stanford University

Topic: Progressive Mesh Parametrization

ACADEMIC SERVICE

Co-Organizer, Toronto Geometry Colloquium <https://toronto-geometry-colloquium.github.io/>

A weekly webseries to promote young researchers and researchers from underrepresented communities.

Reviewer

SIGGRAPH 2024, 2025, SIGGRAPH Asia 2023, 2024, 2025, Transactions on Graphics (TOG) Journal 2025, Transactions on Visualization and Computer Graphics (TVCG) Journal 2023, Eurographics 2021, 2024, 2025, Pacific Graphics 2024, SIGGRAPH Poster 2021, 2025

Panelist

Graduate School Day at SCA 2025, Graduate School Application Seminar at SGI 2022, 2023, HER CODE CAMP 2019

Student Volunteer

SIGGRAPH 2019, SGP 2021

CONTACT INFORMATION

Doug L. James Full Professor, Stanford University, USA

Danny M. Kaufman Principal Scientist, Adobe Research, USA

Alec Jacobson Associate Professor, University of Toronto, Canada

Mathieu Desbrun Advanced Researcher, Inria, France (Previously Full Professor, Caltech, USA)